



Here Be Links: Taming the Data Dragons with OpenAtlas



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Meeting a potential Dragon slayer

- Project site: <https://openatlas.eu>
- Archeological Demo: <https://demo-dev.openatlas.eu/>
- Open-source, browser-based database software
- Designed to acquire, edit and manage research data
- Initiated over 12 ago by Stefan Eichert
- Mainly developed at the [ACDH](#) / [ÖAW](#) and [NHM](#) / [Prae](#)

OpenAtlas Collaborations

- Projects from all fields of the humanities
- Mostly historical, archaeological and prosopographic projects
- A lot of synergies between the projects



THANADOS *RELIC*



MAMEMS

MEDCON



bITEM

APPROACH
BYZANTINUM

Moving Byzantium

DNCCEM

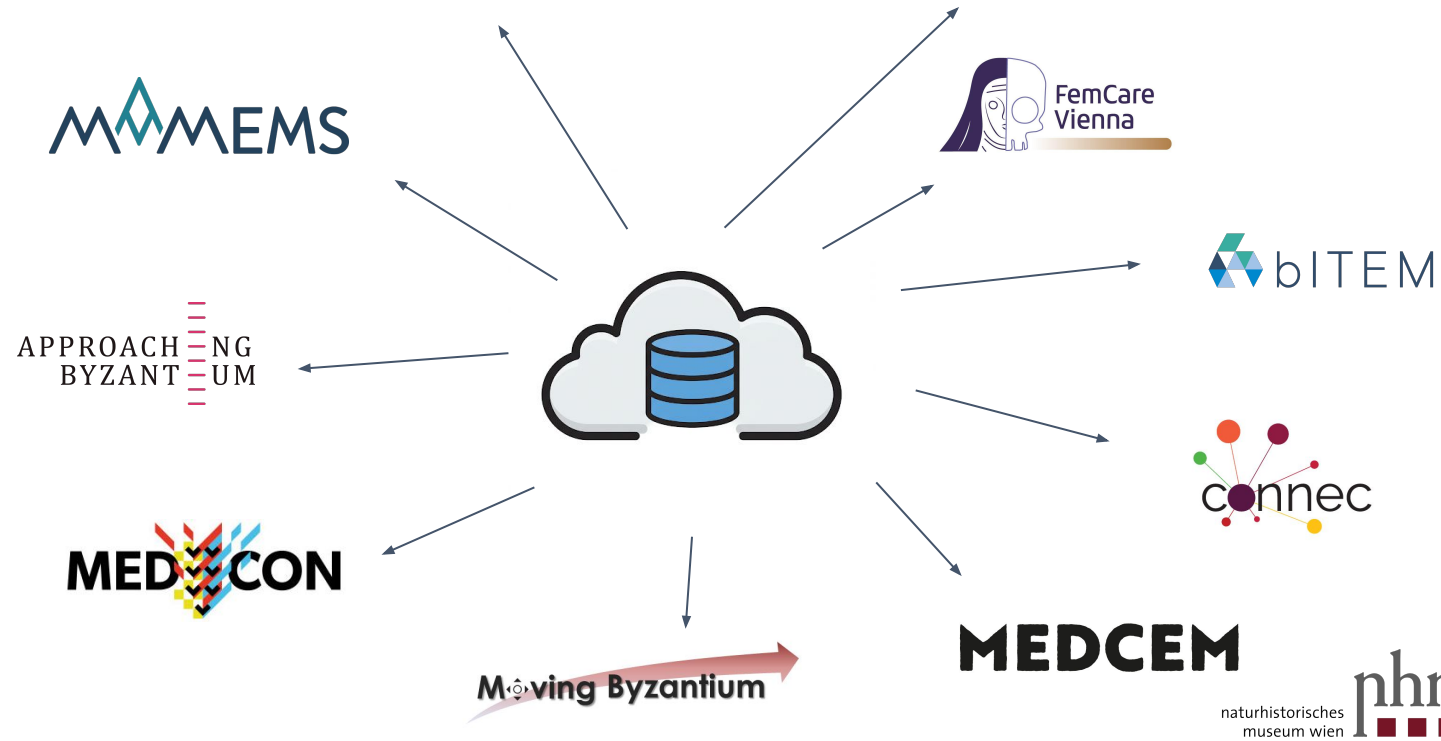
naturhistorisches
museum wien

phm gk



THANADOS

RELIC





MAMEMS

APPROACH
BYZANT
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RELIC

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MEDCEM

DANCEM

THANADOS

naturhistorisches
museum wien

phm

gch



Internet



MAPS of
POWER



Moving Byzantium



Development & Deployment

- Open Source ([GitHub](#)) with monthly release cycles
- Unified codebase: Strictly no project-specific branching
- Quality assurance via [coding standards](#) and automated tests
- [Redmine](#) ticket system & comprehensive [manual](#)
- Debian 13 native, [UV](#), [Docker](#), or [Kubernetes](#)





Taming the dragon?

How we tame the dragon (and partly fail)



Semantic Wall and "Semantic Native" approach

- Semantic Web and CIDOC CRM are terrifying
- Interoperability as a by-product
- "Semantic Native"
- Hiding complexity

From Bones to Books

THANADOS

THANADOS: Spatial analysis of
Early Medieval cemeteries
(Graves, Artifacts, GIS).

RELIC

RELIC: Christianization and
ecclesiastical networks (Texts,
Actors, Concepts).

One shared data model for diverse interdisciplinary demands.



The Spatial Dimension

- Native map integration directly within the browser
- Overlaying excavation plans and drawing geometries natively
- Seamlessly linking finding metadata with spatial data
- GeoPackage (GPKG) export for dedicated GIS analysis



THANADOS

Source Event Actor **Place** Artifact Reference Type File

Search term



Place > Georgenberg > Edit

Thanados 9.2.0



Name *

Alias

Type *

Administrative unit *

Architectural Classification

Case Study

Chronology *

Church Patronage

Country *

Dimensions *

Evidence

Historical place *

Individual Project Types

Individual Project Value Types

JenaTübingen Value Types

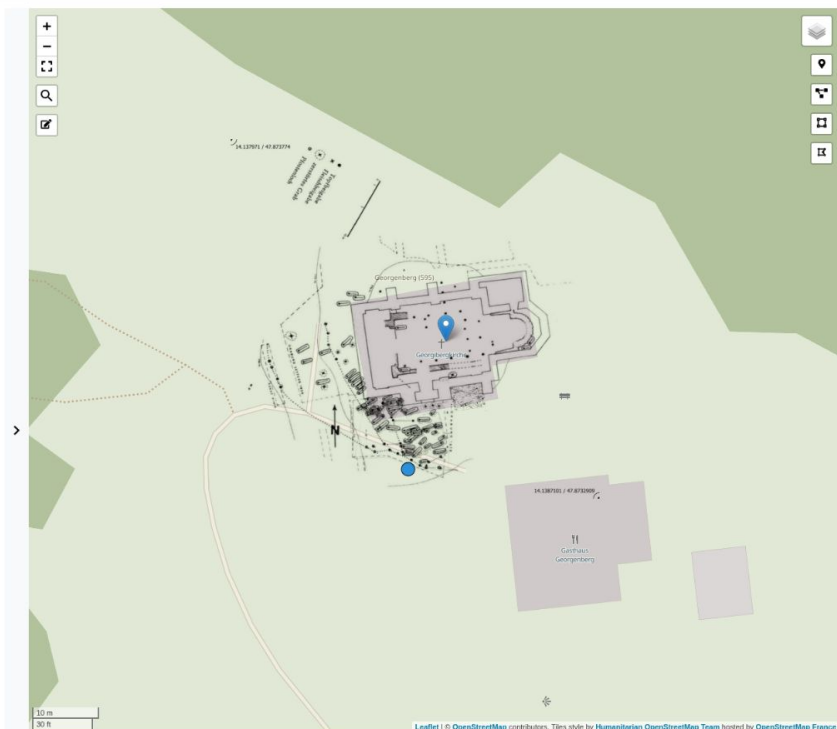
Reference system

Date *

Begin

End

Description



Leaflet | © OpenStreetMap contributors, Tiles style by Humanitarian OpenStreetMap Team hosted by OpenStreetMap France

Here be Links: Linking instead of Importing

- Preserving data quality: No bulk imports from aggregators
- Direct linking to external authorities (Wikidata, VIAF, GND, GeoNames)
- Relationships defined via 'exact match' or 'close match'
- Ongoing development: Seamless UI integration for authority lookups



FAIR enough? Where the Dragon still bites

Findability

- Granular CIDOC CRM metadata and flexible typing enable deep search
- Missing Persistent Identifiers (PIDs) create a risk of "dead links" due to internal database IDs
- Transitioning to UUIDs to provide stable, globally unique URIs
- Global resolution requires institutional infrastructure that transcends single-project lifespans



Accessibility

- Lean tech stack (Debian/PostgreSQL) minimizes technical debt
- Strategic long-term archiving in cooperation with ARCHE
- Open and fully documented REST-API via Swagger
- Format changes currently require URL parameters instead of automated content negotiation
- Deliberate choice against OAuth2/OIDC for API authentication to guarantee absolute data sovereignty



Interoperability

- Native CIDOC CRM structure prevents semantic drift right from the start
- Deep integration of external standards like Linked Places and IIIF
- Missing live SPARQL endpoint for federated queries (actively evaluating Linked Data Fragments)
- Radical approach to external authorities: 100% linking, 0% importing
- Establishing pure semantic bridges (exact match / close match) without replicating external data

Reusability

- Preserving complex historical relationships for deep secondary analysis
- The metadata vacuum: API exports currently lack automated licensing and provenance tags
- Pragmatic attribution: Assigning data ownership via "Case Studies" rather than individual user tracking
- Moving from manual workarounds to UI-driven license mapping
- Integrating with ARCHE to embed explicit CC-BY URIs directly into datasets



Thank you!

OpenAtlas



<https://openatlas.eu/>

<https://thanados.net/>

<https://github.com/craws/OpenAtlas>

<https://demo-dev.openatlas.eu/>

<https://demo-dev.openatlas.eu/swagger>



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